

\$1.00



MSX

MSX & SPECTRAVIDEO COMPUTER CLUB

NEWSLETTER

APRIL

1986

NEXT CLUB MEETINGS (see map below)

APRIL Wednesday 18th 7.30pm

MAY Wednesday 21st 7.30pm

All meetings are at the I.H.C. Hall 56 Ranfurly Rd., Epsom, Auck.
 The club now has a PERMANENT Post Office box at the C.P.O. Auckland.

MSX AND SPECTRAVIDEO COMPUTER CLUB
 P.O. BOX 1733
 AUCKLAND 1, NEW ZEALAND

LIBRARY For copies of software programs or for a list of books & software available write to :- THE LIBRARIAN c/- the club as above.

Club discs are in numbered sets of similar type programs.

COPIES ARE OFFERED ON A NEW TAPE OR NEW DISC WHICH YOU PAY FOR.

Mixtures will not be offered. Basic programs will be on tape and CP/M programs will be on disc. This system is to save time. Lists will

advise names and type of programs in each set. Cost is \$5 each tape, \$12 each disc, payable in advance for postage and library costs.

DONATIONS. All books and articles of interest that you wish to donate to the club library should also be addressed to the librarian.

BORROWING. Books will be available to local members only at club meetings. Out-of-town members may borrow books by post. As books will

now be stored at the club room and accessed once a month please expect a delay in receiving them. Send envelope as above with your

letter. Please return the books after two weeks so that others may use them also. Books will only be sent to members recorded addresses

to assist us in avoiding losses of valuable info.

CONTRIBUTIONS TO THE MAGAZINE Let *

us print that game or article you *

just finished so that others can *

enjoy it too. What can you tell us *

about your pet project? Classified *

ads? Queries? WE NEED YOUR INPUT. *

FORHAT Can be tidy handwritten or *

as typewritten/printer output on *

plain paper, single line spaced, use *

a good black ribbon, use emphasised *

print if possible. Page size should *

be A4 (210 x 297) with margins and *

borders in that size. Send to :- *

The Editor c/- club PO Box. *

Committee members :- If you need *

to phone them please do so before *

9pm, or other reasonable time. *

Chairman Vic 410-8353 *

Secretary Dwayne 278-0755 *

Treasurer David 294-8660 *

Librarian Gary 438-876 *

Ass't Librarian Jack 836-7255 *

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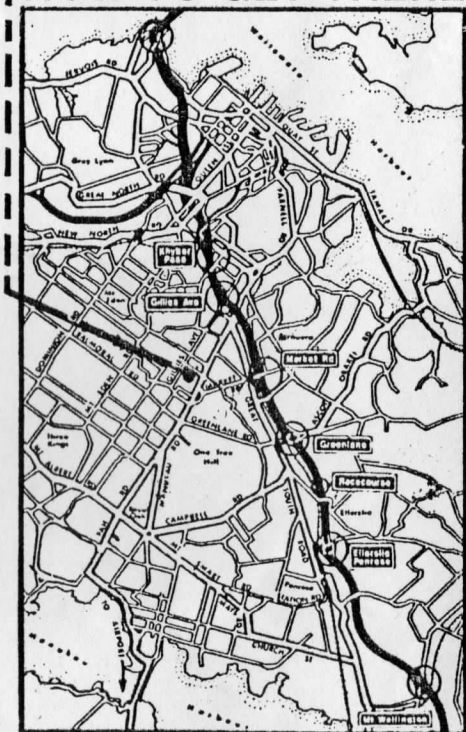
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HOW TO GET THERE





MSX AND SPECTRAVIDEO COMPUTER CLUB

FROM THE EDITOR

HI FOLKS,

Well I've finally managed to scrape our first newsletter together for 1986. There should have been a newsletter in February but there was insufficient material available to publish one and the situation has not improved!! We need regular contributions in order to keep the magazine going and for some of our members this is the only contact they have with the club. The purpose of the magazine is to keep everyone informed of club activities and of what is happening out there in the computing world. It is quite likely you may have seen something in a magazine which others have not read but could be of interest to everyone - we'll never know unless you tell us about it! Maybe you had a problem and have found a solution but aren't really satisfied or maybe you just have problems. Tell us about it and someone just might have an answer to your problem or a better solution to your problem. So stop playing games for a night or two and write something -ANYTHING - for the magazine. ALL CONTRIBUTIONS WILL BE GRATEFULLY RECEIVED AND ALMOST CERTAIN TO BE PUBLISHED.

Just a gentle reminder that we are coming up to our Annual General Meeting and we will be needing several people to fill the committee positions that are becoming vacant. This year some of the people who have served on the committee for the last two years will be standing down. They reckon its time that someone else had a turn. Remember its your club and it needs your enthusiastic support and contributions to survive. Too often people think that they could not do a particular job and then are quite surprised to find that they can do it and actually enjoy it. So give it some thought and give it a go!!!

Have you hired any software/games etc. from the hire club yet? Did you know that SV-318/328 owners can enjoy the MSX cartridge games by hiring the games adapter. I've tried it, it works fine and some of the games are quite superb. If you haven't already done so I recommend you try it out - you won't be disappointed!

At the last meeting there was a modem demonstration and news that the club will have available for hire to club members a modem (and RS232 if required). This will allow you to "tune in" to the local Bulletin board and experience the world of telecomputing. There are some restrictions. You need a SV-328 with disk drive and jack points on your phone and some pills to stop you from falling asleep at the keyboard late at night!

I hope everyone has a happy and prosperous 1986 and let us have a bumper Annual General Meeting!!!

THE EDITOR

FOR SALE

SV 902 SINGLE DISK DRIVE \$500. Contact L.J. CALLENDER, R. D. 3, NEW PLYMOUTH. PHONE (067) 20-760

SV 328 MK1, as new, c/w cassette, reference manuals and six tapes. Only \$550. Also SANYO green monitor \$250. PHONE 658-286. Ask for Rex Butt.

PROBLEM CORNER

One of our readers has constructed a "LISTENING POST" but is having difficulty in getting the software program running. The assembly language program to run the device was written for the Microbee and this would require some modification so that it will run on an MSX computer. The device attaches between the audio output of a short wave receiver and the input port of a computer. It allows the decoding and printing out of morse code, radioteletype (RTTY) and radio facsimile (FAX) pictures. With the device you should be able to 'eavesdrop' on all the weird signals littering the shortwave bands such as international news services, printing your own weather maps and listening on ship-to-shore radio. Is there anyone out there who could help get the software program running and maybe is interested in building such a device for himself. If so contact Steven Balan at telephone 818-4688 (Auck). If you get it going it would be something which that could be published in the magazine!!

Has anyone found out how to access the VIDEO RAM when using the 80-column card. If you have let us know so we can pass it on to others.

Has anyone been able to get the subscript / superscript toggles working properly on their Wordstar.

Does anyone know how to access the cassette drive under CP/M yet?

LIBRARY

Some of the programs available on disk are as follows

Disk 1:

CHEK	COM :	FBA056	COM :	NSWP	DOC :	PCEXPORT	COM
PCFILE	COM :	PCFILE	DOC :	PCSORT	COM :	UNERA15	ASM
UNERA15	COM :	UNERA15	DOC :	UTL#051	DOC		

Disk 2:

LUE	COM :	LU300	INF :	MEX	COM :	MEX	HLP
MODEM	COM :	NSWP	COM :	SVMBASIC	COM		

Disk 3:

AD	COM :	ADVENTUR	DOC :	ADVENTUR	MSG :	ATAB	DAT
COMMON	DAT :	KTAB	DAT :	LTEXT	DAT :	QUEST	BAS
RTEXT	DAT :	STEXT	DAT :	SVDO	COM :	SVDO	MAN
SVINST11	COM :	TRAVEL	DAT :	U-SVDO	COM :	VTERR	DAT

MSX and Spectravideo Computer Club
P.O. Box 1733
Auckland

05-03-1986

Dear Member

The annual general meeting of the MSX and Spectravideo Computer Club will be held on Wednesday April 16. It will start before the regular scheduled club meeting at 7:30 pm, in the I.H.C. Hall, 56 Ranfurly Road.

A number of the committee members will be standing down from their positions, and hence the main business of the evening will be the election of the new committee.

It is important that all members attend this meeting and come along prepared to participate. It's the old story again that all the responsibility has been shouldered by a few and the others have just gone along for the ride.

If the club is to survive members must come forward and contribute something. See you there.

Yours faithfully

D. Smith

Dwayne Smith
Club Secretary

FOR SALE Spectravideo SV-328, Datacassette and more than 100 once-used tape blanks, SV-802 Centronics printer interface. Surplus to requirements. No reasonable offer refused. Phone 686490 or write to Stephen Taylor 49 Wynyard Rd., Auckland 3

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WANTED TO BUY SV-806 80 column card for SV-601/605 expander. Willi Gailer ph 466-236

FOR SALE SV-328 with SV-903 twin track datacassette, amber monitor, SV-605 expander with DS disc drives, 80 col card, SV-3200 130 cps printer, includes all interfaces and heaps of software cp/m and basic. \$2500 ono.

also SV-328 and SV-601 expander, disk-controller & disk drive, \$595 Andrew Hunt ph 586-822

SOME LOCATIONS IN THE RAM MAP

by Ian Macfarlane

My request for information on this subject drew a complete blank, which was discouraging, so this is all my own hackwork.

((Under SV328 disk BASIC from the top of the variable table of your program to the bottom of string space, there is:-

(1) The stack used by basic, starts just below the string space & is usually of limited size, but if you put a program into a tight loop eg. 20 GOSUB 20 then the stack will extend to the top of the variable table then stop with "Out of memory". You'll find the line No and the pointer of that line No repeated.

(2) C100 to C17F:-the 1st half of sector 1, track 0, is the disk bootstrap loader, which can be overwritten without affecting later disk operations.

(3) C000 to C0FF one character set, use unknown.

Above the string area eg D2D8 to D3A0 and the disk buffer area eg D3A1 to F4FF is disk BASIC ie D40F Filemode of currently open file. D403 Filetype of currently open file.

D300 to E07F is the 1st half of each sector from 3 to 17 of track 0. There is a range from D503 to D688 inclusive, that's only used during boot up, and seems a safe place for USR routines. Some entry points beyond are D597:-SAVE, D70A:-LOAD & MERGE, the file allocation table from track 20 sector 15 has been overlayed at DA03 to DB01.

DB00 to E0FF is track 1 in full. Some of the entry points are E03D: NAME, E05D: OPEN, E1E2: KILL, E1FC: SAVE, E393: GEI, E40F: ATTR, E46C: FILES, E6B4: DSK1%, E6F4: DSK0%, EA45: DSKF, EA6D: LOC, EABA: LOF, EA7A: EOF, EABB: FPOS, EB1E to EBFF: Disk error messages, EC02: COPY, and EDCC: IPL.

E500 to F4FF is track 2 sectors 1 to 7, use unknown.

F500 to FFB4 is the working area for BASIC, some locations I've defined are:-

F520 to F53E up to 10 USR routine addresses.

F53F Error No. ERR

F541 Printer carriage position

F542 Printer flag 0=off

F543 screen width, which can be poked from 0 to 40.

F544,2 stacktop (not necessarily current)

F54B,9 Current line No. during run or list (See F7E1,2 also).

F54A,B Program start (see also FDE4)

F54C,D Address beyond the end of a string

F635 to F79D contents of current line entered or listed.

F703 Type prefix ie. No. bytes per current number

F794,7 Temp Program address

F7A2,3 Top of string space

F7C4 No. of bytes in a string

F7C5,6 String start address

F7C7,8 Top of free string space

F7CD,E Address of end of FOR statement

F7CE,DE Line number of a DATA statement

F7D6 Auto flag, 1=on else off.

F7D7,8 Auto line No.

F7D9,A Auto increment.

F7DB,C Program address at the end of previous statement, during run.

F7DD,E Stack pointer at the beginning of each statement.

F7DF,50 Error line No. ERL

F7E1,2 Current line number, entered, being listed or in error.

F7E3,4 Program address at start of line in error.

F7E5,6 CH ERROR subroutine pointer.

F7E7 --FF while in error routine, else 00.

F7EA,D Current break line No.

F7EC,D Program address preceeding "BREAK" line.

F7EE,F Start of variable table.
 F7F0,1 Start of array space.
 F7F2,3 End of array space+1.
 F7F4,5 Address of next DATA to read
 F7F6 to FB0F 26 entries, ie one/letter of alphabet are the variable type prefixes, default values are all 0, ie double precision but are changed by DEFINT A,C-E etc.
 FGF1 Trace flag eg. TRON=AF & TROFF=0.
 F90C Max No. of disc drives
 F98D MAXFILES No.
 F994 Current disc drive number -1
 F997,8 Address of filemode of open file.
 F99D Filename of last disk file used.
 FA02 Click switch 0=off else on. (see also FD74)
 FA03,4 Y & X screen print positions.
 FA05 Cursor flag 0=off 1=on.
 FA06 Screen height(0 to 23)-24.
 FA0A,B,C Foreground, background and border colors.
 FA1E to FABB Current function key definitions.
 FD4C to FD64 Screen line controls, one entry/line (23 lines). 0=line joins next line, AF=ENTER keyed on that line otherwise value =line No
 FD69 to 72 10 FH key flags.
 FD74 Click flag
 FDE4,5 Low memory. & FDE6,7 High memory.=2nd clear param.(defit D5B8)
 For all the following switches, an odd value=on, and even =off.
 FDEB ON KEY(1) switch & FDEC,D =subroutine pointer.
 FDEE ON KEY(2) switch & FDEF,F0 =subroutine pointer.
 FDF1 ON KEY(3) switch & FDF2,3 =subroutine pointer.
 FDF4 ON KEY(4) switch & FDF5,6 =subroutine pointer.
 FDF7 ON KEY(5) switch & FDF8,9 =subroutine pointer.
 FDFA ON KEY(6) switch & FDFB,C =subroutine pointer.
 FDFD ON KEY(7) switch & FDFF,F =subroutine pointer.
 FE00 ON KEY(8) switch & FE01,2 =subroutine pointer.
 FE03 ON KEY(9) switch & FE04,5 =subroutine pointer.
 FE06 ON KEY(10) switch & FE07,8 =subroutine pointer.
 FE09 ON STOP switch & FE0A,B =subroutine pointer.
 FE0C ON SPRITE switch & FE0D,E =subroutine pointer.
 FE0F ON STRIG(0) switch & FE10,1 =subroutine pointer.
 FE12 ON STRIG(1) switch & FE13,4 =subroutine pointer.
 FE15 ON STRIG(2) switch & FE16,7 =subroutine pointer.
 FE18 ON INTERVAL switch & FE19,A =subroutine pointer.
 FE1B ON MDM switch & FE1C,D =subroutine pointer.
 FE2E,F Current time.
 FE30,1 Interval limit.
 FE32,3 Interval counter.
 FE35 0=normal video else inverse video.
 FE37 0=normal cursor else invert cursor.
 FE33 0=saps lock off else on.
 FE36,B Screen No, and 2nd screen parameter (sprite size).
 FE72 to FF01 Hook table. Each hook is 3 bytes and when not in use they are =RET NOP NOP, but in use they =JP 7777 Anyone with a monitor, 80 column card, or 64K RAM card, could help to fill this table in, eg notice the difference with and without the extra device.
 The following are disk hooks FE7F FE82 FE8B FE9A FE9D FEA3 FEA9 FEB5
 FEB9 FEBF FEC7 FECA FEDF FEE3 FEEB FF12 FF18 FF2D FF5

 Brian's comment corner!

X'PRESS owners only. The video processor 9938 is an upgrade of max, to suit max2 (in writing!). It has proved capable of 512 colours but I have not got them controlled yet. A disk upgrade is due later in the year. For existing owners get it from your dealer. Includes 80 cols. under basic. May be able to upgrade VRAM to 128k. Plus other goodies not worked out yet.

```

10 WIDTH40
20 PRINT "This program sets up the function keys"
30 PRINT "to use the SV328 version of ZEN"
40 PRINT
50 PRINT "ZEN is being loaded at $H0000"
60 PRINT "from drive 1":PRINT
70 PRINT "Please wait"
80 CLEAR 300,&H9FFF
90 BLOAD "1:ZEN328"
100 KEY 1,"X=USR(0)" + CHR$(13)
110 KEY 2,"GOTO310" + CHR$(13)
120 KEY 3,"P32" + CHR$(13)
130 KEY 4,"M0A10BH" + CHR$(13)
140 KEY 5,"00C000H" + CHR$(13)
150 KEY 6,"DEFUSR=&H0000:"
160 KEY 7,"CALL
170 KEY 8,"PUSH
180 KEY 9,"POP
190 KEY10,"EQU
200 CLS:PRINT:PRINT "Use F1 to run ZEN"
210 PRINT:PRINT "Use F6 to DEFUSR if required
220 PRINT:PRINT "NOTE A reset of the computer will      (if using a disk based
system)          destroy the contents of C100 to C17F    inclusive which is part
of the ZEN text-file (if one exists)                    MORAL! SAVE it before
saving it!"
230 PRINT "Use F3 key to save a ZEN text file            (from BASIC)
240 PRINT "To reload it to ZEN use BLOAD filename] goto ZEN and find the end of f
ile with F5 and repeated Q commands untill you find the last QD. Add 1 to val
ue then use F4 to alter the end-of-file pointer (in reverse format remember!)"
250 PRINT "(An easier way is to make a note of the last parameter of the BSAVE wh
en you save it)"
260 DEFUSR=&H0000
270 COLOR4,7:END
280 SAVE "1:ZEN":END
290 BSAVE "1:ZEN328",&H0000,&HBC00
300 END:-----
310 A$="1": B$="&H0000":C$="&H"+HEX$(PEEK($H100))+RIGHT$("0"+HEX$(PEEK($H10
0)),2):PRINT "BSAVE"+CHR$(34)+CHR$(34)+", "B$+", "C$"+STRING$(24,29):LINEINPUT
D$:D$=D$+STRING$(ABS(9-LEN(D$)),32):D$=LEFT$(D$,9):BSAVE "1:" +D$,VAL(B$),VAL(C$):E
ND:-----
320 A$="": B$="&H0000":C$="&H"+HEX$(PEEK($H100))+RIGHT$("0"+HEX$(PEEK($H10
0)),2):PRINT "BSAVE"+CHR$(34)+CHR$(34)+", "B$+", "C$"+STRING$(21,29):LINEINPUT
D$:D$=D$+STRING$(ABS(6-LEN(D$)),32):D$=LEFT$(D$,6):BSAVE D$,VAL(B$),VAL(C$):END
330 / If you are running ZEN on a disk system then change the LIMIT at A105 to
D2C7 with modify.
340 / This will restrict your source file to .48k but will protect the DOS.
350 / If you dislike wasting paper then filter out the formfeeds sent to the prin
ter by MODIFYING the following-
360 'A6E2 FE0C CP 00H
370 'A6E4 08 RET Z
380 'A6E5 A7 AND A
390 'A6E6 CD400 CALL 0044H
400 'A6E9 C9 RET
410 / If anyone comes up with a patch that allows the W,V & R commands to work pl
ease let us all know!
420 / FKEYs 1,2 & 6 are used from BASIC
430 / All the others are used from ZEN
440 / Line 320 is the cassette version of line 310. Alter line 110 to suit
450 / Another ZEN patch is to alter the value at A80C, normally 31, to 34/35 for
a 39/40 column screen or 75 for 80 columns.
460 / If you wish to retain the function key display while in ZEN then alter the
byte at A308 to C9.

```



```

470 / If your paper is more (or less) than 66 lines per page then alter the byte
    # A$1C to End of lines -63
480 / After making any patches to ZEN use GOTO 290 to save the new version.(But
    K(11) the source file first)
490 / If you alter THIS program use GOTO 280 to save the new version.
500 / The following listing will give an enhanced printout of the source file (my
    printer ribbon is worn-out) Use GOTO 510
510 A=$HC000
520 A$="":C=0
530 B$=CHR$(PEEK(A+C))
540 A$=A$+B$:C=C+1
550 IF B$=CHR$(13) THEN 570
560 GOTO 530
570 FOR D=1 TO 3
580 LPRINT A$;CHR$(20);'This is a L/F only on my printer
590 NEXT
600 LPRINT
610 A=A+LEN(A$)
620 IF A$="END"+CHR$(13) THEN ENDELSE 520

```

PS. Has anyone got a VIDIOTEX program going yet?

Colin MacRae 4786744

Graham perhaps my experience in obtaining a disk drive may be suitable for the club magazine.

After attending the Sunday workshops, I became convinced that adding a second disk drive could have many advantages. Having purchased a 328 with super expander some time ago, the decision had to be made to either sell the 328 and buy another computer with twin drives, or upgrade the 328. I decided to stay with the 328 and look for the most economical and practical way to change to a twin drive system.

Once the decision was made the options open to me were to either purchase a second disk drive or sell the 601 expander, disc drive, centronics, and 80 column card then purchase a new twin drive as presently supplied by Spectravideo. The disadvantage of buying a single drive would be that each disk drive would be of a different make. The advantage was the price. The TEAC unit I was considering cost approx. \$430 while to sell the 601 with all the bits and pieces would still leave me a balance of between \$900-1100 to find before I could acquire the Spectravideo dual drive system.

At this point in time good fortune favoured me when I was offered two double sided disc drives mounted in a metal case incorporating power supply and cooling fan. This unit was new, and although a little bulky at \$450 who could resist such an offer. This unit is now working very well, and at the time of writing another similar unit is still for sale if anyone is interested.

In the corse of my enquiries some interesting information came to light but not being an expert I cannot guarantee the accuracy of the points listed as follows-

Most makes of 5.25 disk drives can be interchanged.

Double sided disk drives are readily available, while single sided are less popular.

Double sided disk drives are the same price as single.

Track access time of the disk drive cannot be slower than the disk controller. The access time of the disk controller supplied with the 601 is 25m/sec and 12m/sec for the disk drives described above.

The pinout on this Spectravideo disk controller is not a standard configuration, and a circuit board is used at the disk drive end of the ribbon cable to change this to the standard pinout of the drive. Ref. club mag Oct. 1984 page 11.

Power for the Spectravideo disk drive is supplied via the ribbon cable. Apparently this practice is not recommended.

SOUTH AUCKLAND COMPUTERS LTD.

- 214 Great South Road - P.O. Box 720 -

- Papakura - Ph (09)299-6030 -

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the following are additions to the programme library.

Adventure A02

The Colossal cave adventure. Fills almost a whole disk.

P.C.File.

A reasonably good Data base program.

SVDO. A Word Processor that uses the same commands as Wordstar (C). and this one is almost free! only the copying fee.

STRTRK.BAS STARTREK.NOW HOPEFULLY DEBUGGED!

STARTPEK.BAS A different version, it's doc file calls it Supertrek.

All of the above are CP/M so therefore only for disk people.

We now have the following programs on disk and will be able to transfer to tape as and when we can.

futvoy A quiz of your knowledge of the solar system, courtesy of

DISC USERS:- TRY THIS TIP

To produce a list of programmes on your disc, to keep with your disc and its cover, set your printer margins to 32 characters and use the LFILES command. You will get a printout the same as on the 39/40 col screen.

NOTE: All orders to be sent to GND Toys, 1000 North Island, New Zealand. Freight paid to nearest freight depot in the North Island. \$10.00 charge for South Island freight.

worms, viruses, creepers -----

The following are extracts from some articles in Scientific America's Computer Recreation describing a game called CORE WAR where two computer programs in their natural habitat - the memory chips of a digital computer - stalk each other from address to address. Sometimes they go scouting for the enemy; sometimes they lay down a barrage of bombs; sometimes they copy themselves out of danger or stop to repair damage.

CORE WAR was apparently inspired by a story heard some years ago about a mischievous programmer at a large corporate research lab who shall be named X. X wrote an assembly language program called CREEPER that would duplicate itself every time it was run and spread itself from one computer to another within the X corporation. The program had no other function other than to perpetuate itself. Before long there were so many copies of CREEPER that the more useful programs and data were being crowded out. The infestation was not brought under control until someone thought of fighting fire with fire. A second self duplicating program called REAPER was written. Its purpose was to destroy copies of CREEPER until it could find no more and then destroy itself. Apparently REAPER did its job and things were soon back to normal at the X lab.

The story of CREEPER and REAPER seems to be based on a compounding of two actual programs. One program was a computer game called DARWIN, invented by M. Douglas McIlroy of AT&T Bell Labs. The other was called WORM and was written by John F. Shoch of the Xerox Palo Alto Research Center.

In DARWIN each player submits a number of assembly language programs called organisms, which inhabit core memory along with organisms of the other players. The organisms created by one player (and thus belong to the same 'species') attempt to kill those of the other species and occupy their space. The winner of the game is the player whose organisms are the most abundant when time is called. McIlroy invented an 'unkillable' organism. It was immortal but not very aggressive and only won a few games.

WORM was an experimental program designed to make the fullest possible use of minicomputers linked in a network at Xerox. WORM was loaded into quiescent machines by a supervisory program. Its purpose was to take control of the machine and in coordination with WORMS inhabiting other quiescent machines run large application programs in the resulting multiprocessor system. WORM was designed so that anyone who wanted to use one of the occupied machines could readily reclaim it without interfering with the larger job.

The French spy thriller 'Software: La Guerre Douce' is a political fantasy with similarities to the above. The authors Thierry Breton and Dennis Beneich spin a chilling yarn about the purchase by the Soviet Union of an American supercomputer. Instead of blocking the sale the American authorities, displaying studied reluctance, agree to the transaction. The computer has been secretly programmed with a "software bomb". Ostensibly bought to help with weather forecasting over the vast territory of the Soviet Union, the machine, or rather its software, contains a hidden trigger; as soon as the U.S. National Weather Service reports a certain temperature at St. Thomas in the Virgin Islands, the program proceeds to subvert and destroy every

piece of software it can find in the Soviet network.

Inspired by a June 1959 Scientific American article on self reproducing mechanisms by L. S. Penrose, Fredrick G. Stahl of Chesterfield created a miniature linear universe in which humble creatures moved and lived out their destinies. Stahl writes: "Like Core War, I set aside a closed linear segment of main memory in which a creature was simulated by modified machine language. The machine was an IBM Type 650 with drum memory. The creature was programmed to crawl through its universe eating food (non zero words) and creating a duplicate of itself when enough food was accumulated. Like Core War, I had an executive program which kept track of who was alive and allocated execution time among the living creatures. I called it the 'Left Hand of God'". Stahl goes on to discuss his programs ability to reproduce. He also describes an interesting mutation mechanism: a program being copied might experience a small number of random changes in its code. However, Stahl reports: "I abandoned this line of work after one production run in which a sterile mutant ate and killed the only fertile creature in the universe. It was apparent that extraordinary large memories and long computer runs would be needed to achieve any interesting results."

A similar story concerns a game called ANIMAL in which a program tries to determine what animal a human is thinking of by playing a form of Twenty Questions. David D. Clark of the MIT Lab for Computer Science writes that the employees of a certain company devotedly played ANIMAL. While it resembles neither a battle program nor Stahl's simple creatures, ANIMAL achieved the ability to reproduce itself in the corridors of the core through the efforts of a programmer to enhance a key feature of the game. When the program guesses incorrectly what animal the human has in mind, it asks the human to suggest a question it might ask to improve its performance. This feature lead the programmer to invent a certain trick for making sure that everyone always had the same version of ANIMAL. Clark continues: "On a very early computer system, which lacked any shared directory structure, and also any protection tools, the programmer invented a novel way of making the game available to several users. A version of the game existed in one user's directory. Whenever he played the game, the program made a copy of itself into another directory. If that directory had previously contained a copy of the game, the old version was overwritten, which made the game change unexpectedly to the player. If that directory had previously had no version of ANIMAL the game had been offered to yet another user." Clark recalls the ANIMAL was such a popular game that eventually every directory in the company system contained a copy. "Furthermore, as employees of the company were transferred to other divisions they took ANIMAL as well, and thus it spread from machine to machine within the company." The situation eventually became serious so a new "virulent" version was introduced. When this was played it copied itself into other directories not once but twice. Given time all the old versions of ANIMAL would be overwritten. After a year had passed, a certain date triggered each copy of the new ANIMAL program. "Instead of copying itself twice whenever it was invoked, it now played one final game, wished the user 'GOODBYE' and then deleted itself. And thus ANIMAL was purged from the system!!"

The above appeared on a bulletin board and I thought it might be of general interest and might inspire the games writers among us - EDITOR

TECH Character set for your SPECTRAVIDEO

by David McNeill

The following three programs will redefine the characters of your video ram. This is only a temporary change and will be lost by either typing a 'SCREEN 0' command or turning the machine off. It could be useful for entertainment progs. Firstly type in program 1 then save it to disk or a tape as 'TECH'. Then type new. Now type in program 2 and run it. This is to load the data into your m/c. You will have to type in the data listed below (prog 2 can't do it for you!). Now type new again and type in prog 3. This will save your new character set to disk or tape. If using tape, make sure you save the data straight after prog 1. To start 'TECH', simply (c)load TECH and run it!. If you make a mistake, then individually adjust each of the errors using 'VPOKE address,data'. If you need help or further information, phone me at 5344590 or see me at meetings. Don't forget to change the single quotes in the progs to double quotes

program 1

```
10 OPEN '1:tech' FOR INPUT AS #1
20 FOR X=2176 TO 2775
30 INPUT#1,B
40 VPOKE X,B
50 NEXT X:CLOSE#1
```

program 2

```
10 CLS:PRINT 'Enter data from magazine'
20 PRINT 'Address,data'
30 FOR X=2176 TO 2775
40 PRINT X,;:INPUT A
50 VPOKE X,A
60 NEXT X
```

program 3

```
10 OPEN '1:tech' FOR OUTPUT AS #1
20 FOR X=2176 TO 2775
30 B=VPEEK(X)
40 PRINT#1,B:NEXT X
50 CLOSE#1
```

Data for 'TECH'. format is vram address,data

2176 , 248	2296 , 248	2416 , 248	2536 , 112	2656 , 32
2177 , 136	2297 , 136	2417 , 136	2537 , 16	2657 , 32
2178 , 136	2298 , 8	2418 , 136	2538 , 16	2658 , 32
2179 , 200	2299 , 48	2419 , 200	2539 , 16	2659 , 32
2180 , 200	2300 , 48	2420 , 200	2540 , 16	2660 , 96
2181 , 200	2301 , 0	2421 , 200	2541 , 16	2661 , 96
2182 , 248	2302 , 48	2422 , 200	2542 , 112	2662 , 96
2183 , 0	2303 , 0	2423 , 0	2543 , 0	2663 , 0
2184 , 32	2304 , 112	2424 , 248	2544 , 32	2664 , 0
2185 , 32	2305 , 136	2425 , 136	2545 , 80	2665 , 0
2186 , 32	2306 , 8	2426 , 136	2546 , 136	2666 , 248
2187 , 96	2307 , 104	2427 , 200	2547 , 0	2667 , 168
2188 , 96	2308 , 168	2428 , 200	2548 , 0	2668 , 232
2189 , 96	2309 , 168	2429 , 200	2549 , 0	2669 , 232
2190 , 96	2310 , 112	2430 , 248	2550 , 0	2670 , 232
2191 , 0	2311 , 0	2431 , 0	2551 , 0	2671 , 0
2192 , 248	2312 , 112	2432 , 248	2552 , 0	2672 , 0

2193, 136	2313, 80	2433, 136	2553, 0	2673, 0
2194, 8	2314, 80	2434, 136	2554, 0	2674, 120
2195, 248	2315, 248	2435, 248	2555, 0	2675, 72
2196, 192	2316, 200	2436, 192	2556, 0	2676, 104
2197, 192	2317, 200	2437, 192	2557, 0	2677, 104
2198, 248	2318, 200	2438, 192	2558, 248	2678, 104
2199, 0	2319, 0	2439, 0	2559, 0	2679, 0
2200, 240	2320, 240	2440, 248	2560, 64	2680, 0
2201, 144	2321, 144	2441, 136	2561, 32	2681, 0
2202, 16	2322, 144	2442, 136	2562, 16	2682, 120
2203, 56	2323, 248	2443, 136	2563, 0	2683, 72
2204, 24	2324, 200	2444, 136	2564, 0	2684, 104
2205, 152	2325, 200	2445, 184	2565, 0	2685, 104
2206, 248	2326, 248	2446, 248	2566, 0	2686, 120
2207, 0	2327, 0	2447, 0	2567, 0	2687, 0
2208, 240	2328, 248	2448, 240	2568, 0	2688, 0
2209, 144	2329, 136	2449, 144	2569, 0	2689, 0
2210, 144	2330, 128	2450, 144	2570, 120	2690, 120
2211, 144	2331, 192	2451, 248	2571, 72	2691, 72
2212, 248	2332, 192	2452, 200	2572, 104	2692, 72
2213, 48	2333, 200	2453, 200	2573, 104	2693, 120
2214, 48	2334, 248	2454, 200	2574, 124	2694, 96
2215, 0	2335, 0	2455, 0	2575, 0	2695, 96
2216, 248	2336, 248	2456, 248	2576, 64	2696, 0
2217, 128	2337, 136	2457, 136	2577, 64	2697, 0
2218, 128	2338, 136	2458, 128	2578, 120	2698, 120
2219, 248	2339, 200	2459, 248	2579, 72	2699, 72
2220, 24	2340, 200	2460, 24	2580, 104	2700, 72
2221, 152	2341, 200	2461, 152	2581, 104	2701, 120
2222, 248	2342, 248	2462, 248	2582, 120	2702, 24
2223, 0	2343, 0	2463, 0	2583, 0	2703, 24
2224, 248	2344, 248	2464, 248	2584, 0	2704, 0
2225, 136	2345, 128	2465, 32	2585, 0	2705, 0
2226, 128	2346, 128	2466, 32	2586, 120	2706, 120
2227, 248	2347, 248	2467, 48	2587, 72	2707, 72
2228, 152	2348, 192	2468, 48	2588, 96	2708, 96
2229, 152	2349, 192	2469, 48	2589, 104	2709, 96
2230, 248	2350, 248	2470, 48	2590, 120	2710, 96
2231, 0	2351, 0	2471, 0	2591, 0	2711, 0
2232, 248	2352, 248	2472, 136	2592, 8	2712, 0
2233, 8	2353, 128	2473, 136	2593, 8	2713, 0
2234, 8	2354, 128	2474, 136	2594, 120	2714, 120
2235, 24	2355, 248	2475, 200	2595, 72	2715, 64
2236, 24	2356, 192	2476, 200	2596, 104	2716, 120
2237, 24	2357, 192	2477, 200	2597, 104	2717, 24
2238, 24	2358, 192	2478, 248	2598, 120	2718, 120
2239, 0	2359, 0	2479, 0	2599, 0	2719, 0
2240, 112	2360, 248	2480, 200	2600, 0	2720, 32
2241, 80	2361, 136	2481, 200	2601, 0	2721, 32
2242, 80	2362, 128	2482, 200	2602, 120	2722, 240
2243, 248	2363, 216	2483, 200	2603, 72	2723, 32
2244, 152	2364, 200	2484, 80	2604, 120	2724, 96
2245, 152	2365, 200	2485, 80	2605, 96	2725, 96
2246, 248	2366, 248	2486, 112	2606, 120	2726, 96
2247, 0	2367, 0	2487, 0	2607, 0	2727, 0
2248, 248	2368, 136	2488, 168	2608, 56	2728, 0
2249, 136	2369, 136	2489, 168	2609, 40	2729, 0
2250, 136	2370, 136	2490, 168	2610, 112	2730, 72
2251, 248	2371, 248	2491, 232	2611, 32	2731, 72
2252, 24	2372, 200	2492, 232	2612, 48	2732, 104
2253, 152	2373, 200	2493, 232	2613, 48	2733, 104
2254, 248	2374, 200	2494, 248	2614, 48	2734, 120
2255, 0	2375, 0	2495, 0	2615, 0	2735, 0
2256, 0	2376, 32	2496, 80	2616, 0	2736, 0

2257 , 0	2377 , 32	2497 , 80	2617 , 0	2737 , 0
2258 , 32	2378 , 32	2498 , 80	2618 , 120	2738 , 104
2259 , 0	2379 , 48	2499 , 248	2619 , 72	2739 , 104
2260 , 0	2380 , 48	2500 , 200	2620 , 88	2740 , 104
2261 , 32	2381 , 48	2501 , 200	2621 , 120	2741 , 40
2262 , 0	2382 , 48	2502 , 200	2622 , 24	2742 , 16
2263 , 0	2383 , 0	2503 , 0	2623 , 120	2743 , 0
2264 , 0	2384 , 16	2504 , 136	2624 , 64	2744 , 0
2265 , 0	2385 , 16	2505 , 136	2625 , 64	2745 , 0
2266 , 32	2386 , 16	2506 , 136	2626 , 120	2746 , 168
2267 , 0	2387 , 24	2507 , 248	2627 , 72	2747 , 168
2268 , 0	2388 , 152	2508 , 96	2628 , 104	2748 , 232
2269 , 32	2389 , 152	2509 , 96	2629 , 104	2749 , 232
2270 , 32	2390 , 248	2510 , 96	2630 , 104	2750 , 248
2271 , 64	2391 , 0	2511 , 0	2631 , 0	2751 , 0
2272 , 24	2392 , 144	2512 , 248	2632 , 16	2752 , 0
2273 , 48	2393 , 144	2513 , 8	2633 , 0	2753 , 0
2274 , 96	2394 , 144	2514 , 8	2634 , 16	2754 , 80
2275 , 192	2395 , 248	2515 , 112	2635 , 16	2755 , 32
2276 , 96	2396 , 200	2516 , 224	2636 , 48	2756 , 112
2277 , 48	2397 , 200	2517 , 128	2637 , 48	2757 , 104
2278 , 24	2398 , 200	2518 , 248	2638 , 48	2758 , 104
2279 , 0	2399 , 0	2519 , 0	2639 , 0	2759 , 0
2280 , 0	2400 , 128	2520 , 112	2640 , 16	2760 , 0
2281 , 0	2401 , 128	2521 , 64	2641 , 0	2761 , 0
2282 , 248	2402 , 128	2522 , 64	2642 , 16	2762 , 72
2283 , 0	2403 , 192	2523 , 64	2643 , 16	2763 , 72
2284 , 248	2404 , 192	2524 , 64	2644 , 24	2764 , 120
2285 , 0	2405 , 192	2525 , 64	2645 , 24	2765 , 24
2286 , 0	2406 , 248	2526 , 112	2646 , 88	2766 , 24
2287 , 0	2407 , 0	2527 , 0	2647 , 112	2767 , 120
2288 , 192	2408 , 248	2528 , 0	2648 , 32	2768 , 0
2289 , 96	2409 , 168	2529 , 0	2649 , 32	2769 , 0
2290 , 48	2410 , 168	2530 , 128	2650 , 56	2770 , 120
2291 , 24	2411 , 232	2531 , 64	2651 , 40	2771 , 16
2292 , 48	2412 , 232	2532 , 32	2652 , 120	2772 , 96
2293 , 96	2413 , 232	2533 , 16	2653 , 104	2773 , 96
2294 , 192	2414 , 232	2534 , 8	2654 , 104	2774 , 120
2295 , 0	2415 , 0	2535 , 0	2655 , 0	2775 , 0


```

; GENERAL ROUTINES
; Mainly for TEXT screen
; Bob Curwood
; Christchurch Users Group

```

```

ORG 9000H
LOAD 9000H

```

```

; Rom routines

```

```

CLS: EQU 03777H
CHPUT: EQU 00018H
POSIT: EQU 0393EH
FORCLR: EQU 0FA0AH
BAKCLR: EQU 0FA0BH
BORCLR: EQU 0FA0CH
CHGCLR: EQU 03750H
CSRSW: EQU 0FA05H
BREAKX: EQU 0005CH
BEEP: EQU 040BEH

```

```

WMZEH: EQU 0A003H

```

```

JR START

```

```

; To use this PRINT routine
; load DE with the message
; location and CALL PRINT
; The routine first places
; the cursor at the location
; specified by the first 2 #s
; at the start of each message
; 1.e X & Y
; then prints the following
; message which must end with
; a zero. NOTE that linefeeds
; & C/R can also be printed by
; including as E,10,13 outside
; the quotes. See MS4

```

```

; If the cursor positioning is
; not wanted then put a zero
; in both of these positions.

```

```

; Destroys AF & DE

```

```

PRINT: PUSH HL
      EX DE,HL
      LD D,(HL)
      INC HL
      LD E,(HL)
      LD A,D
      OR E
      JR Z,LOOP2
      EX DE,HL
      CALL POSIT
      EX DE,HL
LOOP2: INC HL
      LD A,(HL)
      AND A
      JR Z,EXIT2
      CALL CHPUT
      JP LOOP2
EXIT2: POP HL
      RET

```

```

; COLOUR CHANGE
; To change the foreground,
; background or border,
; load the desired colour(s)
; into FORCLR,BAKCLR,BORCLR
; and CALL CHGCLR.
; Use only 1 to 15 for the
; desired colour and only
; the colour(s) to be changed
; need to be altered
; This example changes the
; colours to blue on cyan (4/7)

```

```

; Destroys AF

```

```

COLOUR: PUSH HL
        PUSH BC
        LD A,4
        LD (FORCLR),A
        LD A,7
        LD (BAKCLR),A
        CALL CHGCLR
        POP BC
        POP HL
        RET

```

```

; CHANGE FOREGROUND COLOUR
; Call with colour in A
; Destroys AF

```

```

FORCHG: PUSH HL
        PUSH BC
        LD (FORCLR),A
        CALL CHGCLR
        POP BC
        POP HL
        RET

```

```

; CHANGE BACKGROUND COLOUR
; Call with colour in A
; Destroys AF

```

```

BAKCHG: PUSH HL
        PUSH BC
        LD (BAKCLR),A
        CALL CHGCLR
        POP BC
        POP HL
        RET

```

```

; DELAY
; This is a simple delay which
; is called with a value in DE
; to give a variable time.
; A count of 1 gives a delay of
; 1/2 second, 1.e 120 gives 1
; min: 600 gives 5 minutes.
; NOTE if you call this routine
; with zero in DE the delay will be
; over 9 HOURS!.
; The CALL BREAKX will jump out of
; the delay if CTRL STOP is pressed

```

```

; Destroys AF,DE
DELAY:    PUSH HL
DEL1:     DEC DE
          LD A,D
          OR E
          JR Z,DELOUT
          LD HL,11411
DEL2:     CALL BREAKX
          JR NC,NOINT
          CALL BEEP
          JR DELOUT
NOINT:    DEC HL
          LD A,H
          OR L
          JR Z,DEL1
          JR DEL2
DELOUT:   POP HL
          RET

```

```

START:    XOR A
          LD (CSRSW),A
          (Turn cursor off)
          CALL CLS
          LD DE,MS1
          CALL PRINT
          LD DE,05
          CALL DELAY
          LD A,1
          CALL FORCHG
          LD DE,MS2
          CALL PRINT
          LD DE,05
          CALL DELAY
          LD A,4
          CALL FORCHG
          LD A,9
          CALL BAKCHG
          LD DE,MS3
          CALL PRINT
          LD DE,5
          CALL DELAY
          LD A,13
          CALL BAKCHG
          LD DE,5
          CALL DELAY
          LD A,15
          CALL BAKCHG
          LD A,1
          LD (CSRSW),A
          (Turn cursor back on)
          LD DE,MS4
          CALL PRINT
          LD DE,10
          CALL DELAY
          CALL BEEP
          CALL COLOUR
          RET

```

```

; Messages
MS1: DB 10,2,"Text Utility",0
MS2: DB 12,5,"Bob Curwood",0
MS3: DB 14,8,"December 1985",0
MS4: DB 0,0,13,10,"This line i"
      DB "s printed without using th"
      DB "e cursor positioning feature"
      DB "re but with a CR/LF before"
      DB " and after to start on a"
      DB " newline",13,10,0
END

```